

Data Structures

2018/2019



Introduction

- What will be covered in this course?
 - We'll study different data structures, the kind of problems for which they are suitable and algorithms with which they are used.



Contacts

- Teaching (Theoretical / Laboratorios)
 - Jorge Barreiros (jmsousa@isec.pt)
 - Notifications and support material is made available through the *moodle*.



Program content

- **1. Introduction**
 - 1.1 - Complexity Analysis
 - 1.2 - collections API
- **2 - Basic Structures**
 - 2.1 Stacks, Queues, Trees
 - 2.2 - Binary Trees
 - 2.3 - Hash Tables
 - 2.4 – Binary *heap*
- **Structures Advanced 3-**
 - 3.1 - *splay Tree*
 - 3.2 – Disjoint Set (other Heaps)
 - 3.3 - Other Structures

In the Laboratory classes algorithms and structures covered in the lectures will be implemented, applied and tested using the Java language.



Evaluation Rules

- **Examination - 10 points**
 - Written exam to be held at the time corresponding evaluation period.
 - Grades from previous seasons greater than or equal to 50% can be reused.
- **Seminar - 1.5 values**
 - Presentation held during term time (**Week 17 to 21 December**) on a topic of your choice selected from a predetermined list.
 - This component of the note is valid for all times evaluation of the academic year, not subject to replacement or improvement.
 - Grades from previous seasons greater than or equal to 50% can be reused.
- **Laboratory Tests - 8.5 values**
 - Performed during the laboratory classes (**Weeks 5-8 of November and 10-14 December**).
 - This component of the note is valid for normal valuation period and optionally other evaluation periods. Outside the normal period, the examination will include an optional practical part which replaces the grade of laboratory tests.
 - Grades from previous seasons greater than or equal to 50% can be reused.
- There is no minimum score in any evaluation component.



Review (Summary)

Normal period	Exam (Ep. Normal) <i>10 values</i>	Seminary <i>1.5 values</i>	Lab tests. <i>8.5 values</i>
Recourse period	Exam (Ep. Rec.) Or Exam (Ép.Normal) if > 50% <i>10 values</i>	Seminary <i>1.5 values</i>	Lab tests. Or Practical test <i>8.5 values</i>
Other periods	Exam (Ép Special.) Or Examination (another time) is > 50% <i>10 values</i>	Seminary <i>1.5 values</i>	Lab tests. Or Practical test <i>8.5 values</i>



Rating - Improvement

- *The exam* grade can be improved by conducting a new exam on the dates established for the purpose.
- The seminary can not be improved.
- The grade of laboratory tests can be **replaced** by the practical part of the exam from the recourse period or later.

Very important:

The improvement of a note released presupposes registration for improvement in academic services.



Bibliography

- **"Data Structures & Problem Solving Using Java"** 3rd Ed., Marc Weiss, Pearson Education
- **"Introduction to Algorithms"** Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, Clifford Stein, The MIT Press; third edition edition (July 31, 2009)

